

The Impact of Dust on Supernova Cosmology

Matt Grayling, Stephen Thorp, Kaisey Mandel

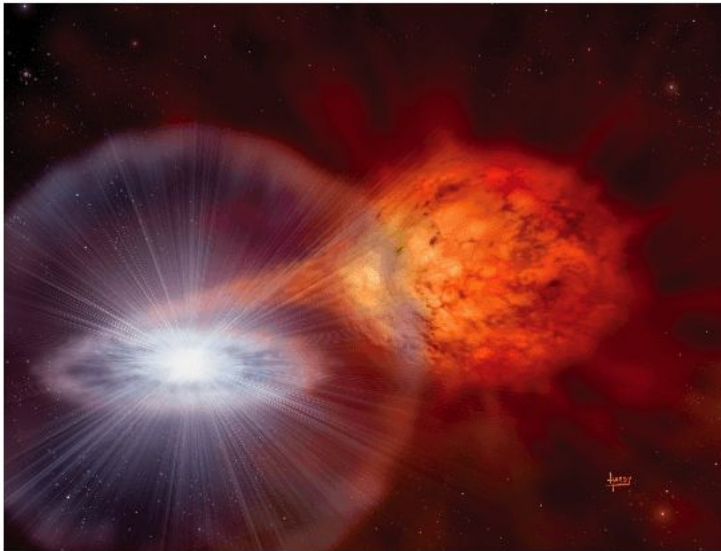


How can we treat dust in a more realistic way?

- Which extinction law should we be using?
- What is the most realistic distribution for A_V ?
- What are sensible priors to use for dust properties?
- Anything else?

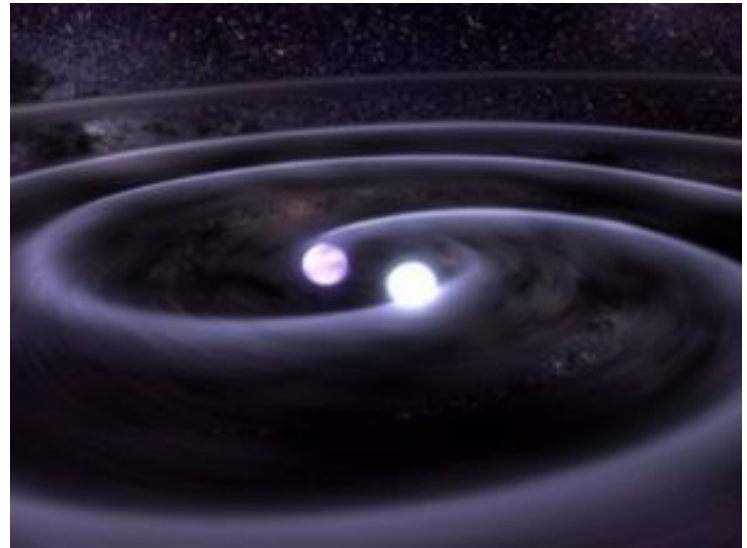
Type Ia Supernovae

Single degenerate



Credit: David A. Hardy, © David A. Hardy/astroasrt.org

Double degenerate



Copyright: GSFC/D. Berry.

$$\mu^s = m_B^s - M + \alpha x_1^s + \beta c^s$$

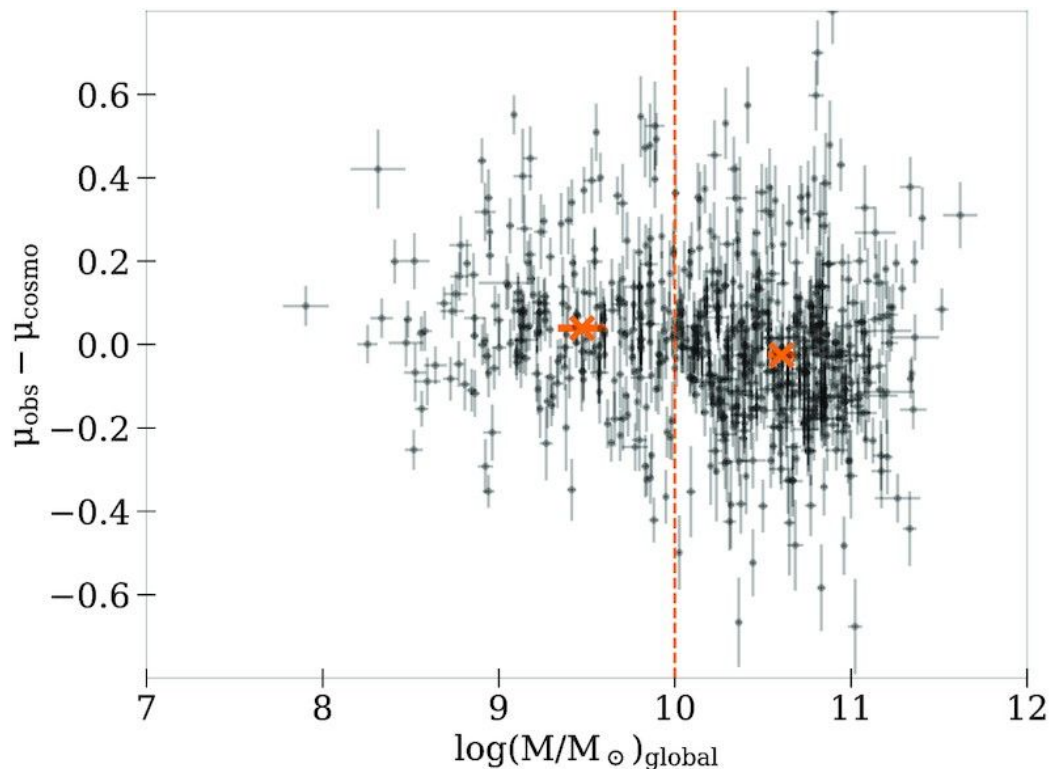
$$\mu^s = m_B^s - M + \alpha x_1^s + \beta c^s$$



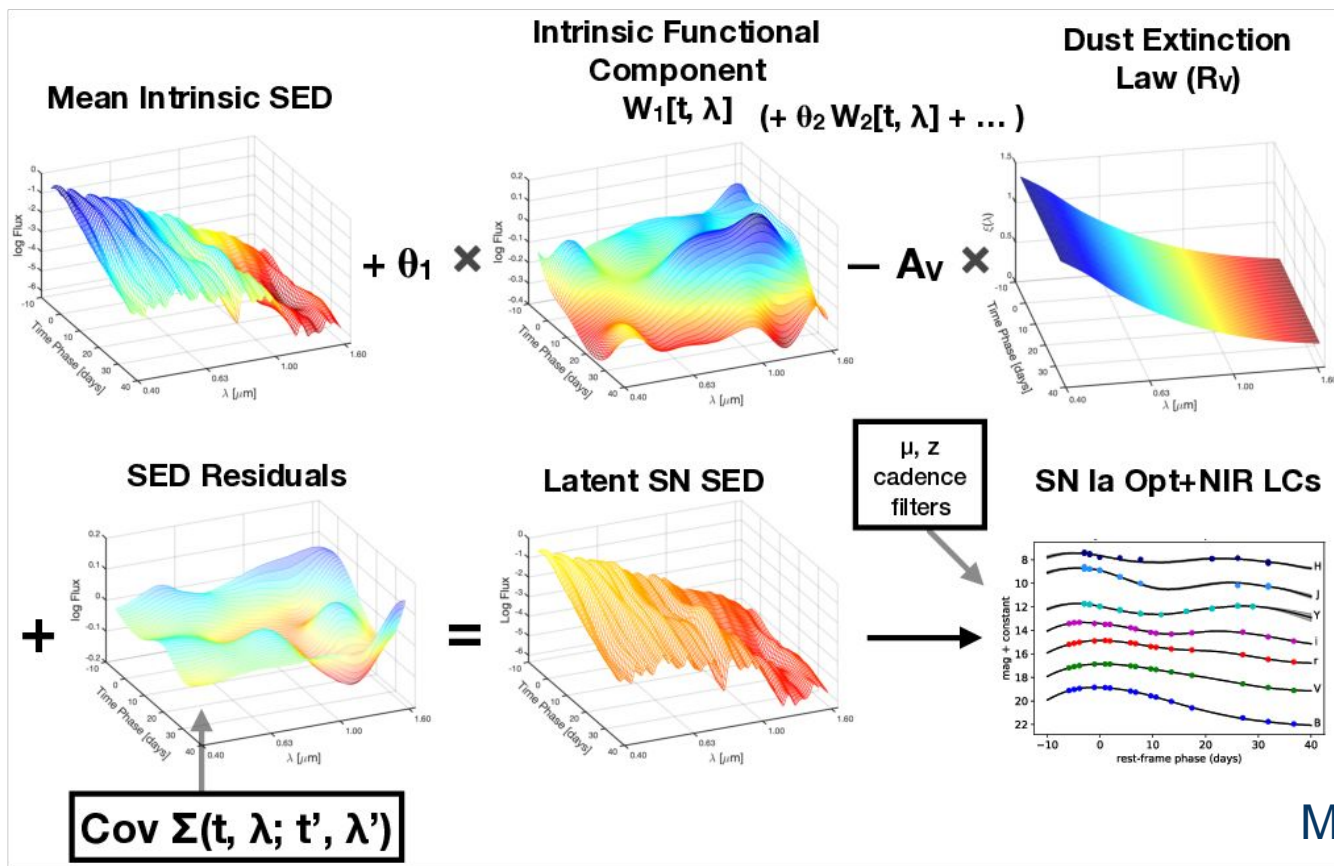
One parameter, two effects

The 'Mass Step'

- Disentangling intrinsic variation from dust vital to understanding host galaxy 'mass step'
- Just different dust properties?
 - SN populations identical, just appear different?
- Or intrinsic differences?
 - Interesting physical implications, different progenitor channels?



The BayeSN Model



- For typical Bayesian inference, each supernova would be fit separately

$$R_V \sim U(1, 5)$$

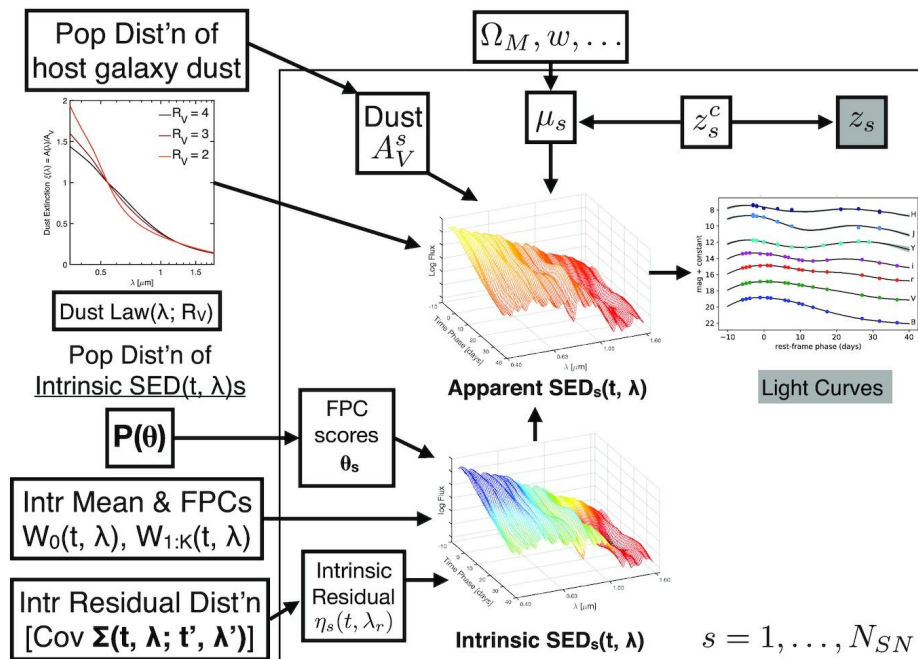
- For a hierarchical model, we jointly sample the posteriors of global and individual supernova properties

$$R_V \sim N(\mu_R, \sigma_R^2)$$

- Stronger constraint from pooling individual objects

Advantages of BayeSN

- Explicit treatment of dust
- Hierarchical Bayesian Model
 - Allows for separate treatment of dust and intrinsic variation across population
- Model extends into NIR wavelengths
 - Better SN Ia standardisation and constraint of host galaxy dust



Mandel+2020

Results: Single Population

Method	μ_R	σ_R	τ_A / mag
Single population	2.58 ± 0.14	0.59 ± 0.20	0.16 ± 0.01

- 475 optical *griz* light curves from $0.015 < z < 0.4$
- With optical only, need to fix cosmology to get reasonable dust constraint
- This is a cosmological sample, objects with $c > 0.3$ have been cut
 - Highly reddened objects are discarded

Results: The Mass Step

Method	ΔM_0 / mag	$\Delta(g - r)_{\text{peak}}^{\text{int}}$	μ_R		τ_A / mag	
			HM	LM	HM	LM
Intrinsic mag diff.	-0.049 ± 0.016	—	2.51 ± 0.16	2.74 ± 0.35	0.18 ± 0.01	0.11 ± 0.01
Intrinsic SED diff.	-0.049 ± 0.027	-0.022 ± 0.010	2.26 ± 0.14	3.36 ± 0.51	0.19 ± 0.02	0.15 ± 0.02

- Our results support the existence of intrinsic differences between SNe Ia in each mass bin
- Depending on our assumptions, our results may also indicate differences in host galaxy dust properties

Results: Dust Evolution with Redshift

Method	$\mu_{R,z0}$	η_R	$\tau_{A,z0} / \text{mag}$	η_τ / mag	σ_R
z evolution	2.58 ± 0.17	-0.38 ± 0.70	0.19 ± 0.02	-0.22 ± 0.06	0.61 ± 0.24

- From $0.015 < z < 0.4$, not that much time for evolution
 - Redshift cuts required to mitigate for selection effects
- Nevertheless, possible redshift evolution needs to be understood

A Plea to the Dust Community

How can we treat dust in a more realistic way?

- Which extinction law should we be using?
- What is the most realistic distribution for A_V ?
- What are sensible priors to use for dust properties?
- Anything else?



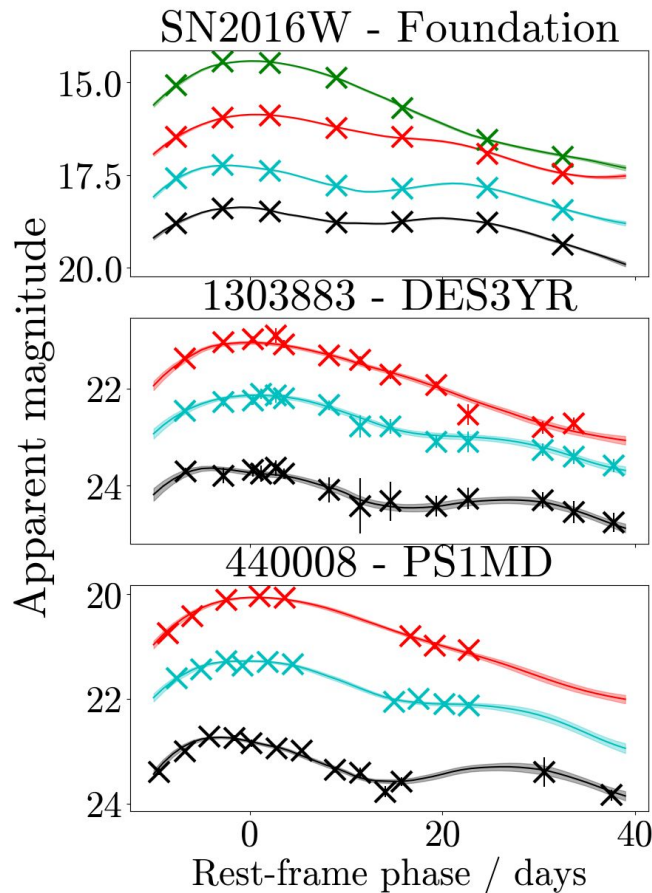
Scan for paper

You can email me at mg2102@cam.ac.uk, in case you have suggestions but have skiing to do...

- Dust is a key source of uncertainty in SN Ia cosmology
- The standardisable nature of SNe Ia allows us to use them to probe dust properties along the line of sight, given a suitable SED model
- We find evidence in favour of intrinsic differences between SNe Ia in different environments - dust may also play a role
- We are as yet unable to constrain a redshift evolution in the R_V distribution of SN Ia host galaxies; understanding this is also vital to cosmology

Investigating the Mass Step

- Infer properties of SN populations separately on each side of the mass step using BayeSN
- Combined sample of 475 SNe Ia from Foundation, DES3YR and PS1MD
- Jointly infer dust and intrinsic differences, see which is more consistent with data



Example
BayeSN
fits

BayeSN Out Now!

New GPU-accelerated, Python implementation of BayeSN out now!

[arXiv:2401.08755](https://arxiv.org/abs/2401.08755), github.com/bayesn/bayesn, pip installable, SNANA integration in progress

 README  MIT license

BayeSN

Hierarchical Bayesian SED modelling of type Ia supernova light curves

Developed and maintained by: Matt Grayling (@mattgrayling), Stephen Thorp (@stevet40), Gautham Narayan (@gnarayan), and Kaisey S. Mandel (@CambridgeAstroStat) on behalf of the BayeSN Team (@bayesn).

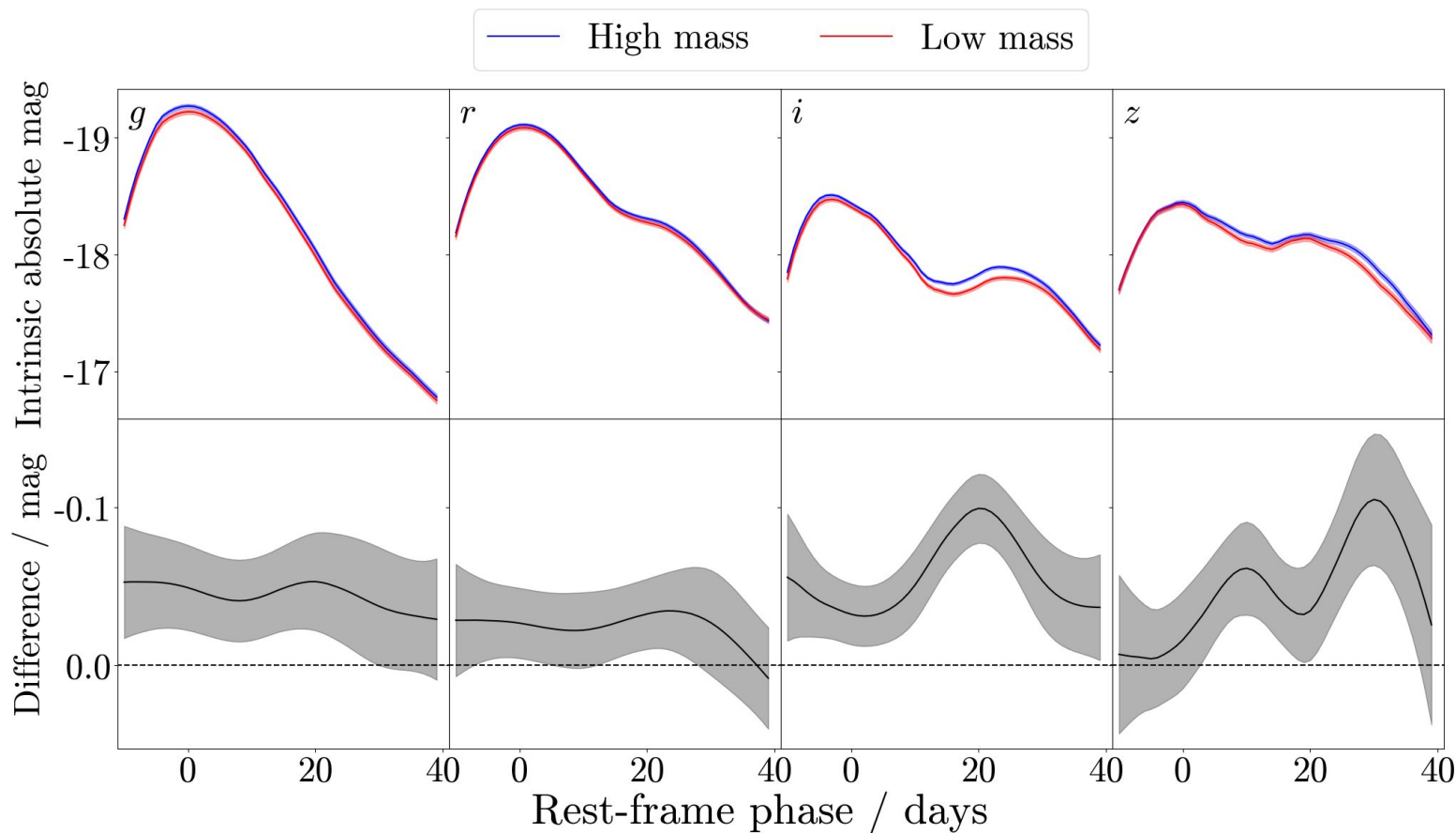
About

BayeSN is a probabilistic optical-NIR SED model for type Ia supernovae, allowing for hierarchical analysis of the population distribution of physical properties as well as cosmology-independent distance estimation for individual SNe. This repository contains an implementation of the BayeSN SED model built with numpyro and jax, with support for GPU acceleration, as discussed in Grayling+2024 (submitted to MNRAS).

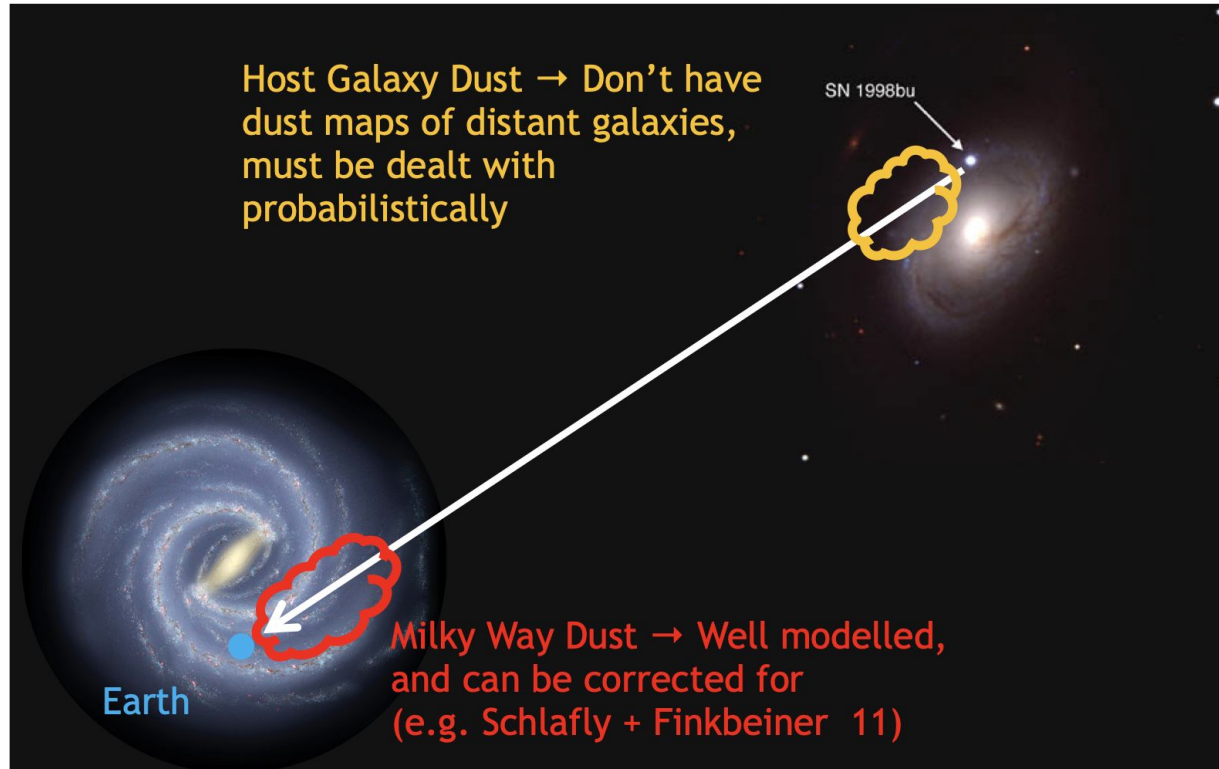


Scan for arXiv link

Intrinsic Baseline Light Curves



Motivation



NASA/JPL-Caltech/ESO/R. Hurt
Nicholas B. Suntzeff

Intrinsic Baseline Colour Curves

